Self-Supervised Deep Learning for Bitemporal Change Detection in Polar Radar Sounder Data

Elena Donini and Francesca Bovolo

Digital Society, Fondazione Bruno Kessler, Via Sommarive, 18 I-38123, Trento, Italy

ABSTRACT

Monitoring the temporal evolution of the cryosphere is essential for understanding the impacts of climate change on polar ice sheets and their contribution to sea-level rise. While remote sensing imagery provides valuable information on surface processes, radar sounders (RS) uniquely enable the investigation of the cryosphere subsurface from the surface to the basal interface. Despite recent advances in machine learning for radargram analysis, existing approaches are limited to single-date acquisitions, and multitemporal analysis remains largely unexplored. This paper presents a novel deep learning method for detecting morphological changes in multitemporal radargrams. The proposed method first learns a baseline representation of subsurface ice structures through a reconstruction network trained on a reference acquisition. During inference, the network extracts deep feature representations from two co-registered radargrams acquired over the same track at different times; Change Vector Analysis (CVA) analyzes these features to detect and characterize changes in the learned feature space. Experiments on co-registered Antarctic MCoRDS-3 radargrams acquired in 2014 and 2016 demonstrate the proposed method's ability to detect and characterize significant stratigraphic changes.

Keywords: Bitemporal change detection, radar sounder, self-supervised deep learning, cryosphere temporal monitoring.

1. INTRODUCTION

Temporal monitoring of the cryosphere is critical to understanding the bidirectional relationship between climate change and the polar ice sheets of Greenland and Antarctica. Rising global temperatures are driving unprecedented ice loss: combined losses from both ice sheets have contributed 22 mm to global mean sea-level rise since 1992, and the rate of mass loss has increased more than fivefold over the satellite era 1. Ice loss, in turn, amplifies climate feedbacks through reduced albedo and freshwater influx into ocean circulation systems 2. A growing body of evidence suggests that even current warming is sufficient to commit several meters of sea-level rise over the coming centuries 2, with abrupt and potentially irreversible changes already underway in both ice sheets 3. Cryosphere monitoring has traditionally relied on optical and SAR remote sensing of the ice surface 4. However, surface observations alone cannot capture the subsurface structure of ice sheets, glaciers, and ice shelves. Radar sounders (RS) address this limitation by providing two-dimensional subsurface profiles (radargrams) that image the cryosphere from the surface to the basal interface. In continental settings, RS enables mapping of englacial stratigraphy, basal topography, and subglacial geology 5. In coastal and marine-terminating regions, RS can characterize the internal structure and thickness of ice tongues, key indicators of ice-shelf stability and calving dynamics 6. Despite the richness of RS data, common methodologies still rely heavily on the visual interpretation of radargrams, a time-consuming and operator-dependent process. Recently, machine learning and deep learning (DL) have begun accelerating information extraction from radargrams, addressing tasks such as ice layer detection, basal interface mapping, semantic segmentation, and unsupervised target detection.[7–13] Nevertheless, these methods operate on single-date acquisitions. The multitemporal analysis of radargrams, which is essential for monitoring change over time, remains largely unexplored, in part because repeated RS acquisitions along the same ground track are rare. Preliminary efforts toward multitemporal RS analysis have addressed: (i) strategies for improving repeated data acquisition 5; (ii) coherent change detection in accumulation radar data to study percolation and refreezing in Greenland firn 14; (iii) differential interferometry (DInSAR) of multipass RS data for vertical displacement estimation of ice layers 6, 15, 16; (iv) non-coherent[17] and coherent[18] multitemporal

E-mails: edonini@fbk.eu, bovolo@fbk.eu

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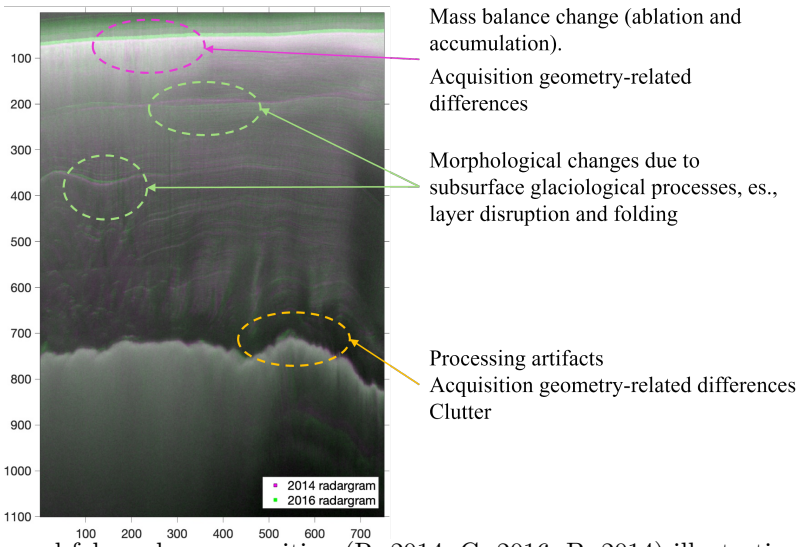


Figure 1: Multitemporal false-color composition (R: 2014, G: 2016, B: 2014) illustrating the different sources of differences observed between bitemporal radargrams: mass-balance and acquisition-related differences near the surface, morphological changes due to subsurface glaciological processes (e.g., layer disruption and folding), and artifacts such as signal dispersion, fading, and clutter.

RS co-registration. However, these studies treat change detection as a secondary objective and do not provide dedicated methods for characterizing temporal changes in radargrams. In contrast, the broader remote sensing community has developed several change detection methods for optical and SAR imagery, achieving remarkable results in Earth surface monitoring. Approaches such as Change Vector Analysis (CVA)[19,20] characterize radiometric change by modeling the expected properties of change vectors (i.e., unchanged samples are expected to show small difference values, while changed samples are expected to show large difference values and different changes in specific directions). More recently, DL-based methods have substantially advanced the state of the art in bitemporal change detection for both optical and SAR imagery 21,22, enabling accurate detection of changes across diverse landscapes. However, to the best of our knowledge, none of these methods has been extended to RS data.

Differences observed between two radargrams acquired over the same track at different times arise from several distinct phenomena, illustrated in Figure 1 on a false-color composition of the 2014 and 2016 acquisitions. Near the surface, differences mainly reflect mass-balance processes (ablation, accumulation, discharge) together with acquisition-related effects. Differences also arise from artifacts, such as signal dispersion and fading, which depend on the specific acquisition and processing chain rather than on the ice itself. Differences may also include morphological changes in the subsurface, such as internal-layer disruption and folding, driven by subsurface glaciological processes. This paper focuses specifically on this last category of change, which is the most directly informative about the evolution of the ice subsurface.

This paper proposes a deep learning-based method to detect and characterize morphological subsurface changes in multitemporal RS radargrams. The core idea is to train a neural network to extract meaningful features from RS data by using the proxy task of reconstructing a baseline radargram acquired at time t_1, minimizing a loss function composed of a Mean Squared Error (MSE) term and a total variation regularization term. At inference, we apply the trained model to co-registered radargrams acquired at both times, t_1 and t_2. We analyze the deep features of the t_1 and t_2 radargrams to select the two most different and informative ones, then process them with Change Vector Analysis (CVA) to identify candidate changes and distinguish change types. Experimental results on a multi-temporal Antarctic dataset acquired in 2014 and 2016 by the MCoRDS-3 sounder demonstrate the effectiveness of the proposed approach. The methodology successfully isolates significant subsurface changes in ice stratigraphy. The rest of the paper is structured as follows: Section 2 proposes the novel methodology for change detection in RS data, Section 3 describes the validation dataset and the experimental

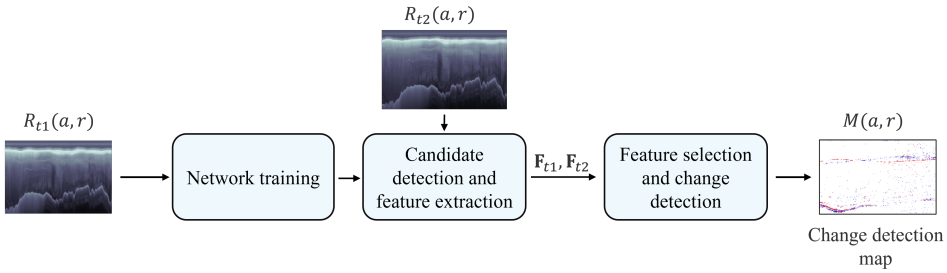


Figure 2: Block scheme of the proposed method for change detection in bitemporal RS data.

setup, and Section 4 presents the experimental results. Finally, Section 5 draws conclusions and presents future directions.

2. PROPOSED METHODOLOGY

Let $\mathbf{R}_{t_1}$ be a radargram expressed in logarithmic (dB) power scale acquired at t_1, defined as a 2D matrix of elements $R_{t_1}(a,r)$, with $r \in \{1,\dots,N_S\}$ indicating the range direction and $a \in \{1,\dots,N_T\}$ indicating the azimuth direction. Let $\mathbf{R}_{t_2}$, of elements $R_{t_2}(a,r)$, $r \in \{1,\dots,N_S\}$, $a \in \{1,\dots,N_T\}$, be a log-scaled power radargram acquired at t_2 on the same track and previously co-registered with methods such as those in 17,18. This paper proposes a novel method for identifying morphological changes in bitemporal RS data. The proposed method consists of three main steps (see Figure 2): (i) training of a deep network to learn the characteristics of the ice subsurface from the radargram acquired at reference time t_1; (ii) computation of the reconstruction residual between $\mathbf{R}_{t_2}$ and its reconstruction $\hat{\mathbf{R}}_{t_2}$ to identify candidate change locations, and extraction and selection of deep features $\mathbf{F}_{t_1}$ and $\mathbf{F}_{t_2}$ from $\mathbf{R}_{t_1}$ and $\mathbf{R}_{t_2}$, respectively; and (iii) analysis of the multitemporal deep features $\mathbf{F}_{t_1}$ and $\mathbf{F}_{t_2}$ to identify K different types of changes $\Omega = \{\omega_1,\dots,\omega_K\}$ and generate the change detection map $M(a,r)$.

2.1 Network training

The first stage of the proposed method learns meaningful subsurface temporal features from the radargram $\mathbf{R}_{t_1}$. We adopt a self-supervised training strategy based on the proxy task of reconstructing the radargram $\mathbf{R}_{t_1}$. We consider a lightweight customized convolutional neural network $f_\theta(\cdot)$ parametrized by θ and designed for radargram analysis. The architecture consists of two parts that have proven effective with RS data (see Figure 3): an Atrous Spatial Pyramid Pooling (ASPP) module[8] to capture subsurface structures at multiple spatial scales, followed by a ShallowNet 7. ASPP applies several parallel atrous (dilated) convolutions to the same input and combines their outputs, enlarging the receptive field and spatial context for each pixel without downsampling. This lets the network capture subsurface structures at multiple spatial scales, from a thin, sharply localized internal reflector to a broad, gradually varying basal-topography trend, within a single feature map. The ShallowNet comprises L convolutional blocks, each consisting of a convolutional layer, batch normalization, and a ReLU activation. The network does not compress information using pooling or other mechanisms, preserving fine details that may indicate possible changes. Despite its limited depth, the network captures the main spatial and radiometric characteristics of radargrams at different scales while preserving the fine-scale structures associated with internal ice stratigraphy. Finally, since neither ASPP nor the convolutional blocks reduce spatial resolution, every intermediate feature map $\mathbf{F}^l$ has the same resolution as the input radargram.

Given the input radargram $\mathbf{R}_{t_1}$, the reconstructed output is defined as:

$$\hat{\mathbf{R}}_{t_1} = f_\theta(\mathbf{R}_{t_1}). \tag{1}$$

The network parameters are optimized by minimizing a loss function composed of a Mean Squared Error (MSE) term and a Total Variation (TV) regularization term:

$$\mathcal{L} = \mathcal{L}_{\text{MSE}} + \lambda \mathcal{L}_{\text{TV}}, \tag{2}$$

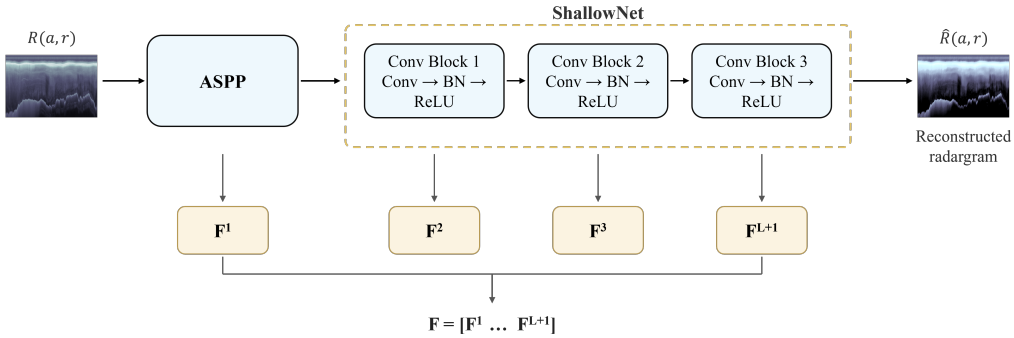


Figure 3: Block scheme of the proposed architecture for change detection in RS data.

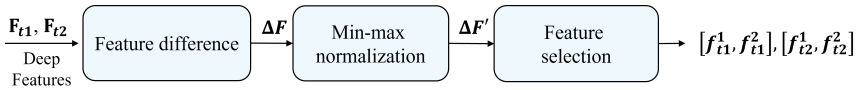


Figure 4: Flow chart of the feature selection.

where λ controls the contribution of the regularization term. The MSE loss is defined as

$$\mathcal{L}_{\text{MSE}} = \frac{1}{N_T N_S} \sum_{a=1}^{N_T} \sum_{r=1}^{N_S} \left(R_{t_1}(a,r) - \hat{R}_{t_1}(a,r) \right)^2 . \tag{3}$$

$\mathcal{L}_{\text{TV}}$ promotes spatial smoothness in the reconstructed radargram:

$$\mathcal{L}_{\text{TV}} = \sum_{a,r} \left| \hat{R}_{t_1}(a+1,r) - \hat{R}_{t_1}(a,r) \right| + \left| \hat{R}_{t_1}(a,r+1) - \hat{R}_{t_1}(a,r) \right| . \tag{4}$$

This regularization reduces the influence of high-frequency noise and encourages the reconstruction of spatially coherent subsurface structures, such as internal ice layers and the bedrock, which are expected to exhibit continuity along the azimuth direction. Once the training is complete, the network $f_\theta(\cdot)$ encodes information about the radiometric and spatial structure of the ice subsurface at t_1.

2.2 Deep Feature Extraction and Selection

After training, $f_\theta(\cdot)$ extracts multiscale features from the co-registered radargrams $\mathbf{R}_{t_1}$ and $\mathbf{R}_{t_2}$. Changes in subsurface structures may appear differently across features: ASPP responds to local radiometric and structural variations with varying degrees of detail, whereas the ShallowNet CNN layers encode increasingly more generalized patterns. We extract features from $L+1$ layers: the ASPP module output and the L convolutional blocks of the ShallowNet. Given a radargram acquired at time $t_k, k \in \{1,2\}$, the feature extracted from layer $l \in L+1$ is denoted as

$$\mathbf{F}^l_{t_k} = f^l_\theta(\mathbf{R}_{t_k}), \tag{5}$$

where $f^l_\theta(\cdot)$ represents the activations of l-th layer. $\mathbf{F}^l_{t_k}$ are concatenated along the channel dimension, yielding:

$$\mathbf{F}_{t_k} = \left[\mathbf{F}^1_{t_k}, \mathbf{F}^2_{t_k}, \ldots, \mathbf{F}^{L+1}_{t_k} \right]. \tag{6}$$

Deep feature extraction produces a large number of features, many of which may be weakly responsive to temporal changes or contain similar information. To address this issue, we introduce a feature selection strategy (see Figure 4). Let $\Delta\mathbf{F}$ denote the temporal feature-difference map between the features extracted from $\mathbf{R}_{t_1}$ and $\mathbf{R}_{t_2}$:

$$\Delta\mathbf{F} = \sqrt{\sum_{c=1}^{N_F} (\mathbf{F}_{t_1} - \mathbf{F}_{t_2})^2}, \tag{7}$$

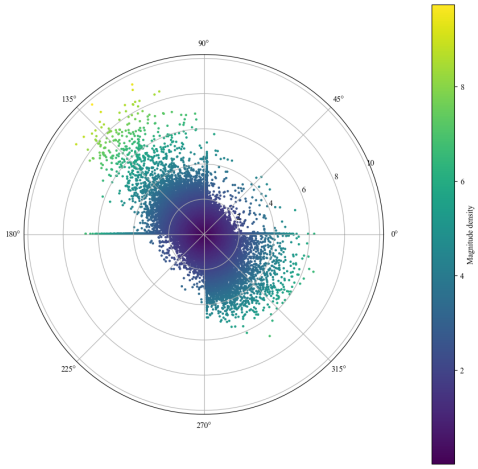


Figure 5: Example of the (ρ, ϕ) probability density distribution.

where N_F indicates the total number of features extracted. Following the assumption introduced in 23, features sensitive to temporal changes are expected to show larger responses in $\Delta\mathbf{F}$, whereas features that primarily encode unchanged pixels should produce smaller variations. We compute the variance of each feature $c \in [1, N_F]$ in $\Delta\mathbf{F}$ and use it as a measure of sensitivity to temporal change. Features are ranked by their variance, with higher-variance features assumed to carry more discriminative information about temporal changes. Conversely, features with low variance are considered less temporally informative and therefore discarded. Although high-variance features are likely to carry change-related information, several may be mutually redundant. To obtain a compact and diverse feature set, as in 10, we perform a second selection stage using the Structural Similarity Index Measure (SSIM) 24. SSIM is computed between pairs of candidate features to assess their similarity. Features with high SSIM values are considered redundant because they capture similar spatial patterns. Therefore, for each group of highly similar features, we retain only the two most representative features $\mathbf{f}^1_{t_k}$ and $\mathbf{f}^2_{t_k}$ with $k = [1, 2]$ and discard the others. The resulting subset consists of complementary features that are highly sensitive to temporal changes and exhibit low structural redundancy.

2.3 Multitemporal Feature Analysis

We analyze the selected features $\mathbf{f}^1_{t_k}$ and $\mathbf{f}^2_{t_k}$ to identify and characterize morphological changes. For each pixel (a, r), we compute the difference magnitude $\rho(a, r)$ as:

$$\rho(a, r) = \sqrt{\sum_{c=[1,2]} (\mathbf{f}^c_{t_1} - \mathbf{f}^c_{t_2})^2}, \tag{8}$$

and a difference direction $\phi(a, r)$ as:

$$\phi(a, r) = \tan^{-1} \frac{\mathbf{f}^1_{t_1} - \mathbf{f}^1_{t_2}}{\mathbf{f}^2_{t_1} - \mathbf{f}^2_{t_2}}. \tag{9}$$

Figure 5 shows an example of the resulting probability density (ρ, ϕ).

In the classical formulation,[20] unchanged pixels concentrate near the origin of the polar plot (a small feature difference yields a small magnitude regardless of direction), while changed pixels form separated clusters at higher magnitude, with ϕ discriminating the type of change.

When dealing with the multitemporal RS data acquired in campaigns operated years apart along a track that is never perfectly overlapping, and under different acquisition conditions (e.g., differing snow/firn/ice temperature, and hence propagation velocity and signal attenuation), a non-negligible difference is observed not only at locations with morphological changes 17, 18, therefore deviating from the ideal situation. Unchanged pixels

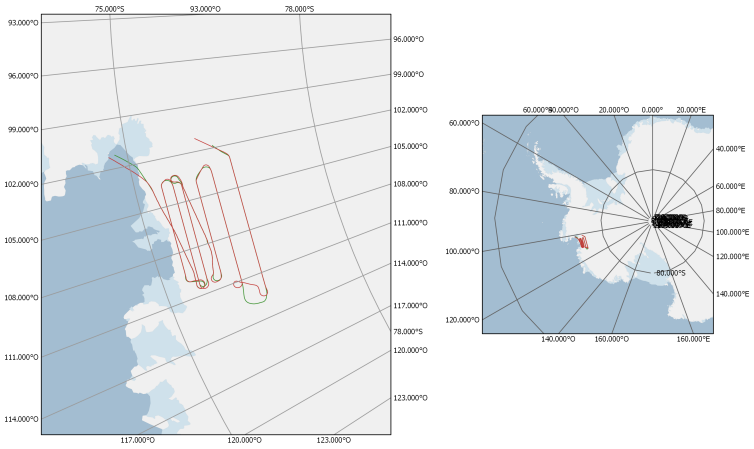


Figure 6: Ground tracks of the campaigns 20141103-04 in red and 20161103-06 in green.

remain clustered near the origin of the (ρ, ϕ) plane, but the boundary with the changed clusters is less sharp (see Figure 5). Candidate change locations are identified by thresholding the magnitude:

$$\rho(a, r) > \tau, \tag{10}$$

where τ is the threshold that separates candidate changed pixels from unchanged ones. Pixels with $\rho(a, r) < \tau$ are labeled background (no change), and the rest are labeled as candidate change locations. Within the candidate locations, we use the direction $\phi(a, r)$ to distinguish change types by isolating K angular sectors. The change detection map $M(a, r)$ assigns each candidate pixel one of K labels based on $\phi(a, r)$, where K varies with the number of morphological changes.

3. DATASET DESCRIPTION

To assess the proposed method, we consider two campaigns that share a similar track in Antarctica and were acquired by the MCoRDS-3 instrument[25] at the Center for Remote Sensing of Ice Sheets (CReSIS). We choose campaigns 20141103-04 and 20161103-06, acquired in 2014 and 2016, respectively, that share most of the track (see Figure 6). Radargrams show i) changes due to glacier movement and snow accumulation during the ice,[18] which are mitigated through co-registration; ii) changes and artifacts due to the different acquisition conditions and post-processing; and iii) morphological changes due to subsurface glaciological processes, such as folding in ice stratigraphy.

We preprocess the data for each campaign, including range compression, azimuth focusing using Synthetic Aperture Radar (SAR), and partial clutter compensation, resulting in the spatial resolutions shown in Table 1. The preprocessing also concatenates adjacent acquisitions and corrects for fluctuations caused by undesired movements during acquisition, such as pitch, roll, and yaw, which could otherwise compromise signal quality. As in 8, we normalize the radargram power by the surface power for each column to inter-calibrate the radargrams and mitigate power fading due to aircraft rolling on curves. For each azimuth column a, we identify the range-sample index of the surface, $s(a)$, and use it to define the corresponding surface power $P_s(a) = \mathbf{R}\big(a, s(a)\big)$. We compared the ground tracks of the two campaigns to identify the along-track segments common to both acquisitions, then used them for feature extraction. We identified four radargrams pairs: three for training and one for feature extraction and computing the change maps (see Table 2 for details). We co-registered the bitemporal radargrams $\mathbf{R}_{t_1}$ from 2014 and $\mathbf{R}_{t_2}$ from 2016 using the non-coherent approach in 17, ensuring spatial alignment between the two acquisitions before feature extraction and change detection.

To quantitatively evaluate the proposed unsupervised method, we selected three regions of interest (ROIs) in RDR4 that exhibit morphological changes: one imaging the surface, one imaging the ice stratigraphy – where layer

Table 1: Parameters of MCoRDS-3[25] for the campaigns acquired in Antarctica in 2014 and 2016. We estimate the radargram geometrical resolutions assuming an ice dielectric permittivity of $\varepsilon_r = 3.15$. AGL stands for above ground level.

Parameters	**Value**
Central frequency (MHz)	190
Wavelength (m)	1.58
Bandwidth (MHz)	50
Sampling Frequency (MHz)	50
Transmitted Power (kW)	6
Altitude AGL (km)	0.5
Range Resolution in ice (m)	2.6
Azimuth Resolution (m)	27.5
Ground sampling distance (m)	15

Table 2: Details of the co-registered radargram pairs used for training (RDR1–RDR3) and testing (RDR4).

Set	ID	Radargram ID start	Radargram ID end	Traces	Samples
Training	RDR1	20141103_04_002	20141103_04_006	16661	2500
		20161103_06_002	20161103_06_006	16666	2500
Training	RDR2	20141103_04_008	20141103_04_010	9996	2500
		20161103_06_008	20161103_06_010	9999	2500
Training	RDR3	20141103_04_016	20141103_04_019	13328	2500
		20161103_06_015	20161103_06_019	16666	2500
Validation	RDR4	20141103_04_012	20141103_04_014	9996	2500
		20161103_06_012	20161103_06_014	9999	2500

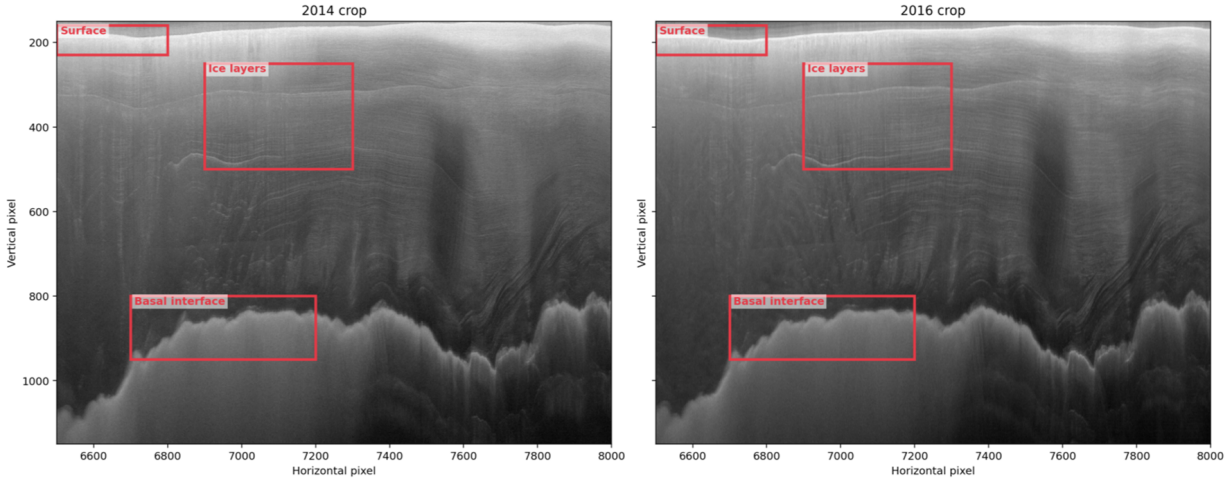


Figure 7: Validation radargram with the highlighted ROI for experimental analysis.

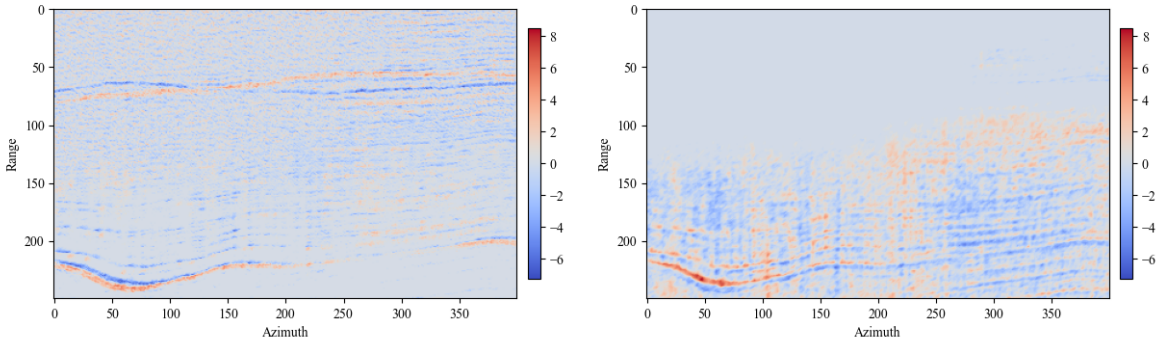


Figure 8: Difference of the selected features $\mathbf{f}_{t1}^1 - \mathbf{f}_{t2}^1$ on the left and $\mathbf{f}_{t1}^2 - \mathbf{f}_{t2}^2$ on the right for analyzing the ice layer ROI.

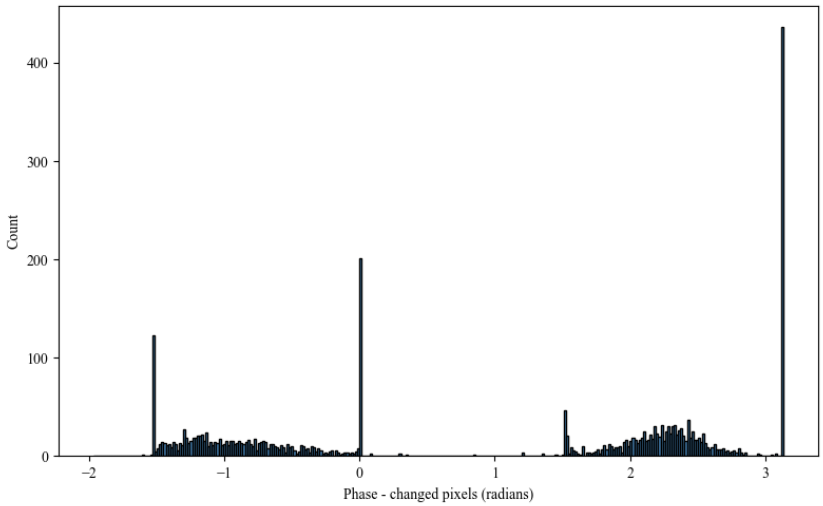


Figure 9: Histogram of $\phi(a, r)$ for changed pixels.

shapes differ between 2014 and 2016 – and one near the basal interface (Figure 7). For each ROI, we produced a reference change map by photo-interpreting the log-ratio between the 2014 and 2016 power radargrams; we use this reference only for a posteriori evaluation of the detected changes, not to train the network or tune the change-detection method, which remains fully unsupervised.

4. EXPERIMENTAL RESULTS

We evaluate the proposed method on the three ROIs for RDR4 (surface, ice stratigraphy, basal interface; Figure 7), both qualitatively by inspecting the resulting change maps and quantitatively against the photo-interpreted reference described in Section 3. Every pixel is assigned to one of three classes: background ω_{nc} (no change, shown in white), ω_1 (backscattering lower at t_2 than at t_1, shown in red), and ω_2 (backscattering higher at t_2 than at t_1, shown in blue).

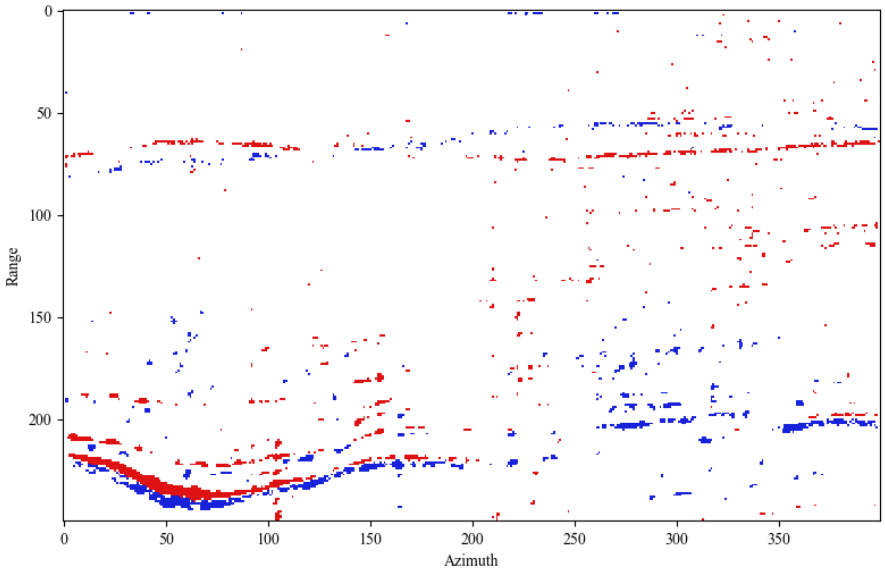


Figure 10: Change map for the ice-layer ROI analyzed in RDR4 (Figure 7), where white indicates ω_{nc}, red indicates ω_1, and blue indicates ω_2.

Figure 8 shows the deep features selected for the ice-stratigraphy ROI. Both respond strongly along a curved reflector at approximately height 180–220 that dips between widths ≈50 and 100 and, more weakly, along a shallower reflector around height 50–90. The opposite-signed responses on either side of these reflectors indicate that the selected features are sensitive to a local shift in the layer position between 2014 and 2016, rather than to a uniform radiometric change across the ROI. Figure 9 shows the distribution of $\phi(a, r)$ for the candidate changed pixels in the ice layer ROI. $\phi(a, r)$ shows two modes, allowing $K = 2$: one mode is centered at -1 rad and the other at 2.5 rad. In the resulting change map (Figure 10), the ice-stratigraphy ROI shows a paired red/blue signature, though with a more continuous structure than two clearly separated bands: a shallower reflector, running at roughly constant range across most of the track, dips locally between azimuth ≈50 and 100, where the red (ω_1) and blue (ω_2) traces are most tightly paired – consistent with a small, systematic deepening of the reflector between 2014 and 2016, with backscatter decreasing where the reflector used to be (ω_1) and increasing at the slightly deeper position to which it moved (ω_2). A second, deeper reflector is visible further down the range axis, most clearly for azimuth <250. Beyond azimuth ≈250, detections become visibly denser and more scattered across both classes, likely reflecting increased speckle or locally lower SNR in that part of the ROI, and a cluster of isolated ω_2 (blue) detections appears near the top of the ROI (low range values) toward the end of the track, which may reflect a near-surface artifact rather than the ice-layer signal of interest.

The surface ROI (Figure 11) shows the same paired signature: a continuous green band (ω_2) immediately above a parallel continuous pink band (ω_1), both tracking the surface reflector's elevation profile across the full ROI width. The vertical separation between the two bands is smallest at the two ends of the track (≈5 pixels) and largest around width 150–280 (≈10 pixels), where the surface profile is also at its deepest, suggesting the apparent shift co-varies with local surface topography (probably related to snow accumulation) rather than being a uniform range offset.

The basal-interface ROI (Figure 12) instead shows a qualitatively different pattern: it is dominated by ω_1 (green), forming a continuous rising ramp from height ≈125 at the start of the ROI to height 10–40 toward its end, with no parallel ω_2 band tracking it as in the other two ROIs, i.e., the 2014 position of the basal reflector is not simply replaced by a nearby, higher-backscatter position in 2016 along most of the track. This is because the radargrams have been co-registered using the bedrock as the reference reflection, and the identified change

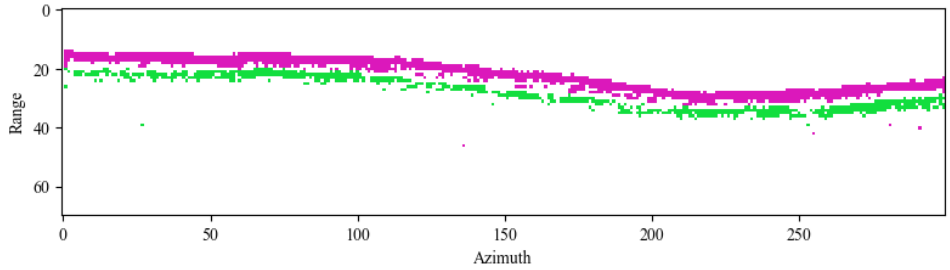


Figure 11: Change map obtained for the surface ROIs analyzed in RDR4 (Figure 7), where white represents no change, pink represents ω_1, and green represents ω_2.

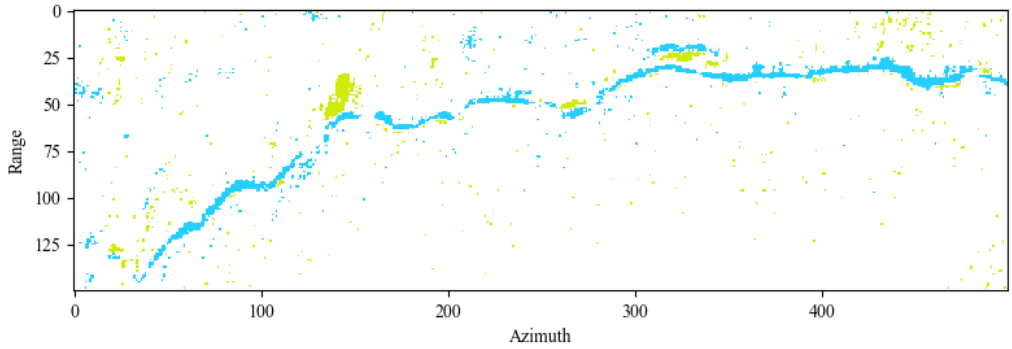


Figure 12: Change maps for the basal interface ROIs in RDR4 (Figure 7), where white represents ω_{nc}, green represents ω_1, and light blue represents ω_2.

reflects different signal dispersion in the range direction, which may be related to different acquisition conditions. Finally, a spatially localized cluster of ω_2 (light blue) appears around a width of 130–160 and a height of 30–60, which may be related to a processing or acquisition artifact.

Table 3 reports the per-class accuracy obtained for the three ROIs against the photo-interpreted reference. The background class is recognized with high, comparable accuracy across all three ROIs ($\geq$0.97), consistent with its being the majority class in every ROI. Accuracy on the two change classes is lower (0.73–0.80), but ω_2 is detected more accurately than ω_1 in all three ROIs, by 0.070 for the ice-stratigraphy ROI, 0.032 for the surface ROI, and 0.059 for the basal-interface ROI. This holds even for the basal-interface ROI, whose spatial pattern (a single dominant ramp plus a localized cluster) differs qualitatively from the paired red/blue signature of the other two ROIs. Averaged across the three classes, the ice layers, surface, and basal-interface ROIs achieve an overall accuracy of 0.834, 0.839, and 0.838, respectively.

Table 3: Per-class accuracy against the photo-interpreted reference of Section 3, for the three ROIs of Figure 7. Mean is the unweighted average of the three per-class values.

ROI	Background	ω_1	ω_2	Mean
Ice stratigraphy	0.9705	0.7315	0.8012	0.834
Surface	0.9705	0.7572	0.7896	0.839
Basal interface	0.9730	0.7407	0.7999	0.838

Across all three ROIs, the two change classes are detected along spatially coherent, geophysically plausible structures – the ice-layer reflectors, the surface profile, and the basal interface – rather than as noise scattered uniformly across the ROI, supporting the feasibility of the proposed feature-based change detection approach for characterizing morphological changes, such as layer disruption and basal-interface shape changes, in multitemporal radar-sounder data.

5. CONCLUSIONS AND FUTURE WORK

This paper presented a self-supervised deep learning method for bitemporal change detection in radar sounder data, combining a reconstruction-based feature extractor trained solely on a reference acquisition with the bitemporal feature differences. Applied to three regions of interest (surface, ice layers, and basal interface) extracted from co-registered Antarctic MCoRDS-3 radargrams acquired in 2014 and 2016, the method detected changes along spatially coherent, geophysically plausible structures rather than as noise scattered across the ROIs, and achieved per-class accuracies of 0.73–0.80 for the two change classes against a photo-interpreted reference, alongside background accuracy above 0.97 in all three ROIs. The qualitative analysis further revealed a consistent paired change-class signature in the surface and ice-layer ROIs, suggestive of a small, systematic shift of the corresponding reflectors between the two acquisitions, and a qualitatively different, spatially localized signature in the basal-interface ROI, consistent with an actual local change in basal topography rather than a uniform effect. These results indicate that deep features learned purely self-supervised from a single acquisition can support unsupervised detection and characterization of subsurface morphological changes, without requiring labeled multitemporal training data, which is important given how rarely repeated radar-sounder acquisitions over the same track are available.

The evaluation was limited to three manually selected ROIs within a single radargram. As future work, we plan to extend it to full-track, multi-site evaluation and to additional MCoRDS-3 campaigns beyond the 2014–2016 pair, thereby allowing a more comprehensive assessment of the method's generalization. Moreover, the apparent systematic reflector shift observed in the surface and ice-stratigraphy ROIs warrants independent verification, for instance against co-registration residuals or auxiliary elevation/accumulation data, to confirm whether it reflects a genuine glaciological change or a residual geometric effect of the co-registration procedure.

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